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PII: S0963-9969(24)00416-2
DOI: <https://doi.org/10.1016/j.foodres.2024.114346>
Reference: FRIN 114346

To appear in: *Food Research International*

Received Date: 1 December 2023
Revised Date: 2 April 2024
Accepted Date: 17 April 2024



Please cite this article as: Tieghi, H., de Almeida Pereira, L., Viana, G.S., Katchborian, A., Santana, D.B., Mincato, R.L., Dias, D.F., de Paula, D.A.C., Soares, M.G., de Araújo, W.G., Bueno, P.C.P., Effects of geographical origin and post-harvesting processing on the bioactive compounds and sensory quality of Brazilian specialty coffee beans, *Food Research International* (2024), doi: <https://doi.org/10.1016/j.foodres.2024.114346>

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Effects of geographical origin and post-harvesting processing on the bioactive compounds and sensory quality of Brazilian specialty coffee beans

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Highlights

Target metabolite profiling supported the characterization of specialty coffee beans

The geographical origin influenced specialty coffee's chemical profile and quality

Post-harvesting processing can enhance the content of health-beneficial compounds

Abstract

Specialty coffee beans are those produced, processed, and characterized following the highest quality standards, toward delivering a superior final product. Environmental, climatic, genetic, and processing factors greatly influence the green beans' chemical profile, which reflects on the quality and pricing. The present study focuses on the assessment of eight major health-beneficial bioactive compounds in green coffee beans aiming to underscore the influence of the geographical origin and post-harvesting processing on the quality of the final beverage. For that, we examined the non-volatile chemical profile of specialty *Coffea arabica* beans from Minas Gerais state, Brazil. It included samples from Cerrado (Savannah), and Matas de Minas and Sul de Minas (Atlantic Forest) regions, produced by two post-harvesting processing practices. Trigonelline, theobromine, theophylline, chlorogenic acid derivatives, caffeine, caffeic acid, ferulic acid, and *p*-coumaric acid were quantified in the green beans by high-performance liquid chromatography with diode array detection. Additionally, all samples were roasted and subjected to sensory analysis for coffee grading. Principal component analysis suggested that Cerrado samples tended to set apart from the other geographical locations. Those samples also exhibited higher levels of trigonelline as confirmed by two-way ANOVA analysis. Samples subjected to de-pulping processing showed improved chemical composition and sensory score. Those pulped coffees displayed 5.8% more chlorogenic acid derivatives, with an enhancement of 1.5% in the sensory score compared to unprocessed counterparts. Multivariate logistic regression analysis pointed out altitude, ferulic acid, *p*-coumaric acid, sweetness, and acidity as predictors distinguishing specialty coffee beans obtained by the two post-harvest processing. These findings demonstrate the influence of regional growth conditions and post-harvest treatments on the chemical and sensory quality of coffee. In summary, the present study underscores the value of integrating target metabolite analysis with statistical tools to augment the characterization

of specialty coffee beans, offering novel insights for quality assessment with a focus on their bioactive compounds.

Keywords: *Coffea arabica*; HPLC-UV-DAD; Specialty coffee; Multivariate data analysis

1. Introduction

Coffee, along with tea, is one of the most consumed and traded beverages in the world, with significant global economic importance (Kwon et al., 2015; Alcantara, Dresch & Melchert 2021; Zarebska et al., 2022). The species *Coffea arabica* (arabica coffee) and *Coffea canephora* (robusta coffee), Rubiaceae, are responsible for almost the entire global coffee production, accounting for around 70% and 30% of the market, respectively (Alcantara, Dresch & Melchert 2021; Hall, Trevisan & De Vos, 2022). As a commodity, coffee is cultivated mostly in subtropical regions including African, Asian, Pacific, Caribbean, Central, and South American countries (Giraud et al., 2019; Hall, Trevisan & De Vos, 2022; ICO, 2023). Brazil is the main producer and exporter, holding more than one-third of the world's production (MAPA, 2023). In the country, the state of Minas Gerais is the largest producer, with large cultivation areas in four main producing regions, namely Sul de Minas, Cerrado de Minas, Chapada de Minas and Matas de Minas (ABIC, 2021).

In 2022, Brazil exported approximately 2.2 million tons of coffee to 145 countries. In the same year, coffee exports (in the form of green coffee, soluble coffee, and extracts) reached a value of US\$ 9.2 billion due to the high price of the product on the international market (CONAB, 2023). Worldwide, the trading of greater added-value products is being boosted by the specialty coffee segment. Based on a demanding market and novel processing methods targeting high quality and distinct sensory attributes, specialty coffees emerged as a key player capable of providing differentiated products, reflecting greater growth rates than ordinary coffees (Malta et al., 2021).

Specialty coffees are those produced following standardized protocols, including cultivation, selection, and preparation, which reflects on the final beverage sensory profile (Ogut et al., 2022; Várady et al., 2022). After roasting, the resulting aroma, flavor, and other sensory attributes are decisive criteria for consumer preference, product scoring, and, consequently, pricing (Agnoletti et al., 2022). To calculate the quality score, the main method uses a standardized sensory evaluation protocol performed by Q-graders, following the procedures developed by the Specialty Coffee Association (SCA). It is considered specialty coffee those who achieve a score equal to or greater than 80 (Zarebska et al., 2022).

However, the distinctive quality linked to flavor and aroma is dependent on the chemical composition of the green beans (Giraud et al., 2019; Hall, Trevisan & De Vos, 2022). Green beans consist mainly of carbohydrates, proteins, lipids, amino acids, peptides, organic acids, alkaloids, and phenolics (Várady et al., 2022). After roasting, these compounds will release volatiles such as alcohols, acids, esters, aldehydes, ketones, and furans, and develop the characteristic aroma and flavor in the cup, in addition to other important sensory attributes such as body, acidity, and sweetness (Malta et al., 2021; Várady et al., 2022). More than 1000 volatile compounds are generally detected in roasted coffee, and their concentrations may

vary according to the different roasting parameters, storage time, and packing conditions (Pereira et al., 2019; Hall, Trevisan & De Vos, 2022). Therefore, evaluating only the volatile compounds in the roasted beans may make it difficult to compare coffees obtained at different times, from different locations and suppliers.

Nevertheless, both green and roasted coffee beans are rich in non-volatile bioactive compounds, most of them showing psychostimulant and antioxidant properties, among others (Várady et al., 2022; Febrianto & Zhu, 2023). The main bioactive compounds in coffee include alkaloidal and phenolic compounds. Diterpenes such as cafestol and kahweol also stand out among important bioactive compounds in coffee beans with potential pharmacological actions such as anti-inflammatory, hepatoprotective, anti-cancer, anti-diabetic, and anti-osteoclastogenesis activities (Ren et al., 2019; Febrianto & Zhu, 2023; Strocchi et al., 2023).

The chemical composition of green beans as well as the accumulation of non-volatile compounds such as caffeine, trigonelline, chlorogenic acids (CGA), and sucrose is highly influenced by the geographical origin and environmental growing conditions (Taveira et al., 2014; Kwon et al., 2015), and storage conditions (Zarebska et al., 2022). Besides, different post-harvesting processing practices including basic processing methods (natural, washing, and semi-washing) and new innovative processing methods (anaerobic fermentation and carbonic maceration) also influence the raw bean chemical composition and, consequently, the final beverage quality after roasting (Pereira et al., 2019; Várady et al., 2022; Febrianto & Zhu, 2023). Considering that the coffee cup quality is greatly influenced by the pre- and post-harvesting processing, modulating coffee aroma and tasting remains challenging (Velásquez & Banchón, 2023).

Undoubtedly, much attention has been paid to the post-harvesting processing and roasting parameters, aiming to increase the coffee cup quality. However, the factors determining the influence of the chemistry of the initial green bean in the final cup have been less studied, although they are the starting point for every coffee product (Hall, Trevisan & De Vos, 2022).

Given the significant interaction among the genetic background, geographical origin, and post-harvesting procedures in defining the uniqueness of specialty coffee (Alcantara, Dresch & Melchert 2021; Hall, Trevisan & De Vos, 2022; Várady et al., 2022), this study delves into the comparative chemical composition analysis of specialty green coffee beans from three specific regions within the Brazilian state of Minas Gerais. The impact of two post-harvesting processing methods (pulp and natural drying) on the chemical profile of such samples was also investigated. The research further illustrates how this composition aligns with sensory classification and offers chemical benchmarks for further assessment for quality assurance and origin verification. For that, well-established and accessible target analysis based on high-performance liquid chromatography coupled to diode array detection (HPLC-UV-DAD) was employed to quantify eight major phenolic compounds and alkaloids biomarkers including trigonelline, theobromine, theophylline, chlorogenic acid derivatives (expressed in 3-caffeoylquinic acid), caffeine, caffeic acid, ferulic acid, and *p*-coumaric acid in the green beans. After roasting, all samples were subjected to sensory analysis for coffee grading. Therefore, the aims of the present study were (1) to ascertain the impact of the two main post-harvesting processing methods on the sensory score and chemical profile of specialty coffee

beans, and (2) to explore the potential of chemical profiles as indicators of regional origin for both processed and unprocessed beans.

2. Material and Methods

2.1. Chemicals and reagents

For the chromatographic analyses, HPLC grade acetonitrile (99.8%), methanol (99.8%), and formic acid ($\geq 98.0\%$) were purchased from Sigma-Aldrich (St Louis, MO, USA). Ultra-high purity water was prepared by filtration using a Milli-Q system (Milli-Q-Plus Direct-Q 3 UV-R) from Millipore (Bedford, MA, USA). The standards trigonelline, theophylline, theobromine, caffeine, chlorogenic acid, caffeic acid, ferulic acid, *p*-coumaric acid, and naringenin were purchased from Sigma-Aldrich (St Louis, MO, USA).

2.2. Green coffee beans

Green beans of specialty coffee (*C. arabica*) were obtained from three different regions of Minas Gerais state including Cerrado, Matas de Minas, and Sul de Minas, comprising samples from different altitudes, ranging from 700 to 1370 m (Table 1 and Table S1, Supplementary Information). All samples were harvested and processed in 2021 with the curation of a technician specialist of the Technical Assistance and Rural Extension Company of Minas Gerais State (Emater-MG). The experimental design included a total of 78 samples, 39 corresponding to natural cherries (directly dried without any processing, NAT) and 39 pulped cherries (subjected to de-pulping before drying, PRO) (Table 1). Detailed information including metadata is summarized in Table S1 (Supplementary Information). The green and intact grains were packed in sealed plastic bags and stored at 10 °C until sample grinding. For that, 10 g of the green beans were ground into powder using a Hamilton Beach mill (Hamilton Beach Brands, Inc., USA), and the particle size was standardized using a granulometric sieve (20 mesh). Finally, 100 mg (± 2 mg) of each sample powder was transferred to 2.0 mL Eppendorf tubes and stored at -20 °C until sample extraction and analysis.

2.3. Sample preparation

In the present study, pulped coffees (PRO) were produced by the removal of the skin and pulp of the ripe cherries manually or using electrical pulpers before the drying process. The natural processing method (NAT) was performed by simply drying the whole ripe cherries (containing the mucilage, pulp, and bean) without further manipulation. In both processing methods, the drying was carried out under the sun, combined or not with mechanical drying for approximately 15 days, depending on the place where the samples were obtained. In any case, the final humidity was kept at around 10-12%, as recommended by the usual post-harvesting practices (Velásquez & Banchón, 2023). After gridding, each

sample ($100 \text{ mg} \pm 2 \text{ mg}$ of green coffee powder weighed in 2.0 mL Eppendorf tubes) was extracted with 1 mL of an extraction solution (SE) composed of methanol 80% v/v (MeOH:H₂O 8:2) containing 200 $\mu\text{g/mL}$ of naringenin (internal standard, IS). All samples were subjected to ultrasound extraction for 3 cycles of 6 minutes each, with manual agitation between each cycle. After extraction, samples were centrifuged at $2800 \times g$ for 5 minutes (centrifuge model 5804 R, Eppendorf, Hamburg, Germany). An aliquot of 600 μL of the supernatant was filtered through a 0.22 μm PTFE filter (1.5 cm of diameter) directly to a 1.5 mL Eppendorf tube. Then, 250 μL of each filtrate was transferred to a 1.5 mL glass vial, containing 500 μL of the extraction solution (SE) for subsequent chromatographic analysis.

2.4. High-performance liquid chromatography analysis (HPLC-UV-DAD)

The chromatographic analysis was performed using a Shimadzu liquid chromatograph (Kyoto, Japan) equipped with two LC-20AT pumps, a SIL-20A autosampler, a CTO-20AC column oven, an SPD-M20A photodiode array detector, and a SCL-10A vp controller, operated by a LC Solution 1.23 SP1 software. The chromatographic separations were performed using a C18 chromatographic column of $150 \times 4.6 \text{ mm i.d.}$, and a particle size of 5 μm (Phenomenex, Torrance, CA, USA), protected by a guard column of the same stationary phase and supplier. The mobile phase consisted of purified water acidified with 0.2% of formic acid (v/v), pH 2.7 (solvent A), and methanol containing 10% of acetonitrile and 0.2% of formic acid (v/v) (solvent B). The optimized elution gradient ranged from 10% to 25% of B, from 0 to 2 minutes, then from 25% to 50% of B from 2 to 16 minutes, and from 50% to 100% of B until 22 minutes. After that, 100% of B was kept for 6 minutes and the elution program came back to 10% of B in 1 min. Additional 7 minutes at 10% of B guaranteed column re-equilibration. The flow rate was 1.0 mL/min and the injection volume was 10 μL . Detection of analytes was performed at a wavelength window ranging from 200 to 800 nm. Peaks were assigned by comparison with authentic standards retention time and UV spectra, of both the standards and samples, under the same analytical conditions. For the quantitative analysis, peaks were integrated at 268 nm for trigonelline, 275 nm for theophylline, theobromine, and caffeine, and 325 nm for chlorogenic acid, caffeic acid, ferulic acid, and *p*-coumaric acid (Table 2). Naringenin, used as an internal standard, was integrated at 275 nm.

2.5. Quantification of coffee bioactive compounds

Calibration curves of trigonelline, theophylline, theobromine, caffeine, chlorogenic acid (precisely 3-caffeoylquinic acid or 3-CQA, representing all caffeoylquinic acids), caffeic acid, ferulic acid, and *p*-coumaric acid were elaborated in concentrations ranging from 25 to 200% of the expected concentration of each compound in the green coffee samples ($n = 3$). Initially, stock solutions (SS-1) of each compound were prepared in methanol 80% v/v (MeOH:H₂O 8:2) at 10 mg/mL, except for theobromine, which was prepared at 2.5 mg/mL. Then, a second stock solution (SS-2) containing all standards was prepared and sequentially diluted to achieve the final concentrations required for the analytical curves. For that, solution SS-2 was prepared by transferring aliquots of 280 μL , 16 μL , 2 μL , 250 μL , 500 μL ,

30 μL , 20 μL and 40 μL of each SS-1 solution of trigonelline, theobromine, theophylline, caffeine, chlorogenic acid, caffeic acid, ferulic acid and *p*-coumaric acid, respectively, into another 2.0 mL Eppendorf tube. The solvent was then evaporated in a SpeedVac (Concentrator plus System, Eppendorf, Hamburg, Germany) for 5 hours. Finally, the residue was resuspended with 1.0 mL of methanol 80% v/v containing 200 $\mu\text{g/mL}$ of naringenin (IS), resulting in a solution containing all the standards at the appropriate concentrations for preparing the analytical curves. Finally, aliquots of 25 μL , 50 μL , 100 μL , 150 μL and 200 μL of solution SS-2 were transferred to 1.5 mL Eppendorf tubes, and the final volume of 1 mL was achieved by transferring 975 μL , 950 μL , 900 μL , 850 μL and 800 μL of methanol 80% v/v containing 200 $\mu\text{g/mL}$ of naringenin, respectively. Table S2 (Supplementary Information) displays the concentrations achieved for each compound quantified in the present study.

Calibration regression curves ($y = ax + b$) were made by plotting the peak-area ratio between the analyte area and the IS area (y) against analyte concentration in $\mu\text{g/mL}$ (x). The final concentration of each compound in the green coffee beans was calculated considering the dilution factor (= 3) and the weighted mass. The results were expressed in mg/g. Additionally, for each compound, the limit of detection (LOD) and limit of quantitation (LOQ) of the optimized method were calculated using the calibration curve parameters. For that, $3.3\sigma/S$ and $10\sigma/S$ were used to calculate LOD and LOQ, respectively, where S is the slope and σ is the standard error of the intercept of the calibration curve equation (Table S3, Supplementary Information).

2.6. Sensory analysis

The sensory evaluation was made in collaboration with the Technical Assistance and Rural Extension Company of Minas Gerais (Emater-MG) during the 18^o Minas Gerais Coffee Quality Contest in 2021. The procedure followed the protocols described by the Specialty Coffee Association – SCA (SCA, 2008). For that, 100 g of the green beans of each sample were roasted until reaching the color standard #65 for whole beans, according to the Agron/SCA Color Classification System, respecting the time range of 8 to 12 minutes. The roasted beans were ground so that 70-75% of the powder passed through a 20-mesh sieve. For cupping, the beverage was prepared at the proportion of 8.25 g of coffee powder per 150 mL of water at 93 °C, and the sensory analysis was performed by professional Q-graders trained and certified by the SCA. The brewed coffee samples were tasted between 8 to 10 minutes after the preparation, at an approximate temperature of 70 °C. Seven Q-graders evaluated three cups of each sample, making a total of seven repetitions per sample. The 10 attributes defined in the protocols described by the SCA (SCA, 2008) were evaluated: fragrance/aroma (aromatic aspects of the dry and infused coffee), uniformity (consistency of flavor of the different cups tasted), clean cup (lack of interfering negative impressions from first ingestion to final aftertaste), sweetness (perception of the presence of certain carbohydrates), flavor (coffee's principal character, as a combined impression of all the gustatory sensations and retro-nasal aromas), acidity (often described as "brightness" when favorable or "sour" when unfavorable), body (the tactile feeling of the liquid in the mouth), balance (the various aspects of flavor, aftertaste, acidity and body of the sample working

together, complementing or contrasting each other), aftertaste (the length of positive taste and aroma qualities), and overall impression (meant to reflect the holistically integrated rating of the sample as perceived by the individual panelist). These attributes were graded on a 10-point scale with intervals of 0.25 points. The final sensory score was generated from the sum of the evaluated attributes and averaged considering the seven evaluators' grades. All tested samples had a final score higher than 85 points (Table S1, Supplementary Information).

2.7. Statistical analysis

The data matrix containing the quantitative analysis results (mg/g) and the metadata were organized using Microsoft® Excel® for Microsoft 365 MSO (v.2305 Build 16.0.16501.20074). Average, standard deviation, and Student's t-test (95% of significance) were calculated using the same Microsoft® Excel® version. Descriptive statistical analysis, normality test, two-way ANOVA, Tukey's test, and correlation analysis were performed using Prism (version 8.0.2, GraphPad software©, La Jolla, CA, USA). For the multivariate data analysis, the data was normalized against sample weight, mean-centered, and subjected to principal component analysis (PCA) using the online software MetaboAnalyst 5.0 (Montreal, QC, Canada) (<https://www.metaboanalyst.ca/>) (Xia et al., 2009; Pang et al., 2021).

To discern whether the processing type (NAT or PRO) could be predicted from a set of geographical, chemical, and sensory variables, Multivariate Logistic Regression (MLR) analysis was employed. The binary processing type served as the dependent variable, while independent variables included altitude, targeted metabolite concentrations, and sensory scores, detailed in Tables S1 and S4 (Supplementary information). The model's efficacy was quantified by odds ratios (ORs) with 95% confidence intervals (CIs). Pseudo- R^2 values and overall model classification accuracy were calculated to assess the proportion of processing type variance explained by the included predictors. The model's predictive power was further evaluated through Receiver Operating Characteristic (ROC) curve analysis, with the Area Under the Curve (AUC) measuring its classification accuracy as NAT or PRO. The results were visually represented in an ROC plot contrasting model sensitivity and specificity.

Complementing the MLR, univariate statistical analyses including Spearman's rank correlation and Mann-Whitney U/Student's t-tests addressed the main associations and group differences. Given the non-normal distribution of most variables, Spearman's correlation was the preferred method for assessing relationships. The choice between non-parametric Mann-Whitney U and parametric Student's t-tests for comparing groups was dictated by the data distribution, verified by a suite of normality tests including the Anderson-Darling, D'Agostino & Pearson, Shapiro-Wilk, and Kolmogorov-Smirnov. MLR and univariate statistical analyses were performed using the Prism software (version 10.0.2, GraphPad software©, La Jolla, CA, USA).

3. Results and discussion

3.1. Target analysis of phenolic and alkaloidal bioactive markers

Accurate information about the relative chemical composition of the major bioactive compounds in green coffee beans can be obtained by several analytical techniques. When associated with chemometric tools, it is possible to differentiate a large number of samples from different origins, varieties, or treatments, and identify the contributing factors to the observed clustering patterns (Alcantara, Dresch & Melchert 2021).

Advancements in understanding coffee chemistry have been achieved from studies employing sophisticated and powerful techniques such as nuclear magnetic resonance (^1H NMR) (Kwon et al., 2015; Muniz et al., 2023), fluorescence spectroscopy (FE) (Botelho et al., 2017), near-infrared spectroscopy (NIR) (Giraud et al., 2021; Ribeiro, Salva & Silvarolla, 2021), and high-performance liquid chromatography coupled to diode array detection and/or mass spectrometry (HPLC-DAD-MS/MS) (Babova et al., 2016; Várady et al., 2022; Ogutu et al., 2022; Zarebska et al., 2022). Among these, chromatographic methods are particularly advantageous due to their ability to separate, detect, and accurately quantify compounds even in complex matrices (Bueno & Lopes, 2020). Moreover, chromatography provides distinctive chemical fingerprints that facilitate the differentiation and authentication of food and/or plant materials by pinpointing minimal pattern differences or identifying exclusive chemical markers (Alcantara, Dresch & Melchert 2021).

Considering its widespread adoption in chemical analysis, for being one of the most worldwide consolidated techniques, and widely disseminated in chemical laboratories, the present study employed HPLC-UV-DAD for the separation, detection, and quantification of eight bioactive chemical markers in green coffee beans. The normalized results were followed by statistical and multivariate data analysis. The chromatographic method was optimized to separate the target compounds within a 22-minute elution gradient (Figure 1, Table 2). We analyzed a set of 78 coffee samples using a non-exhaustive extraction method in three replicates ($n = 3$). All peaks' identities were assigned with the aid of authentic reference standards, which also provided the calibration points for the quantitative analysis (Table S3, Supplementary information). All analytical curves showed a linear correlation coefficient (R) above 0.996 ensuring their fitness for quantitative analysis.

3.2 Relative chemical composition and geographical background

One of the goals of the present study was to ascertain if the bioactive chemical composition of green coffee beans, whether processed or not, can be associated with their region of origin. For that, principal component analysis (PCA) was used to investigate the relationships between the chemical composition and their respective region of origin. The samples of specialty coffee beans obtained by the two different processing methods (NAT and PRO) were analyzed separately. The data was normalized by the weighted mass and internal standard, and mean-centered previously to the PCA. The score plots of both NAT and PRO beans indicated that the first three principal components captured a significant

amount of variability in the dataset, accounting for 97.7% and 92.4% of the explained variance, respectively.

The score plot corresponding to the NAT coffee beans from Cerrado shows a subtle trend of cluster separation, underlying differences in the concentration of the targeted metabolites (Figure 2B). The plot suggests a possible distinction between samples from the Cerrado and Sul de Minas regions. The samples from Matas de Minas overlap with the other two regions, indicating that their chemical profile presents intermediate characteristics with those of Sul de Minas and Cerrado. Matas de Minas and Sul de Minas are part of the same Atlantic Forest biome (Figure 2A), sharing a phytoecological background that may explain their higher intersection in the PCA plot. On the other hand, the overlap with the Cerrado samples may be due to similarities in latitude, which influences the climate, including temperature and solar irradiance.

The same subtle trend can be observed in the pulped coffee beans (PRO). However, the de-pulping process seems to homogenize differences among beans from Sul de Minas and Matas de Minas regions, as indicated by Figure 2C. Also, it is interesting to point out that processed coffee beans from Cerrado displayed less variability compared to the non-processed beans, as evidenced by the more compact Cerrado clustering. Although the limitations on sample size and imbalance between the groups, these PCA outcomes shed light on the hypothesis that green coffee beans' chemical composition can be indicative of their geographical origin. The high percentage of variance explained for both processing types confirms the robustness of the developed PCA model in effectively summarizing the main patterns presented in the data.

Therefore, it is worthwhile to mention that the level of overlap between clusters indicates that a higher number of samples and high statistical power are required to detect significant differences between the chemical profiles of samples from each region. It is especially important since the lower number of Cerrado samples also does not allow a deeper examination into other environmental factors, such as water availability, soil erodibility, or altitude (Figure S1 and Text S1, Supplementary information).

Following the multivariate analysis, two-way ANOVA and Tukey test were employed to examine the geographic influence on the average concentration of enriched biomarkers in green coffee beans, processed or not (Figure 3). For that, the compounds were enriched according to the KEGG (Kyoto Encyclopedia of Genes and Genomes) database and divided into four main classes according to their biosynthetic pathways (Figure S2, Supplementary information). This enrichment step and, consequently, the grouping of these eight compounds into four main classes, was important to understand the role of such compounds in a biosynthetic pathway level. For that reason, we performed the enrichment according to the KEGG database to ensure the correct classification. Therefore, for the statistical tests, it was used the sum of all xanthine derivatives (theobromine, theophylline, and caffeine) and the sum of all cinnamic acid derivatives (ferulic acid, *p*-coumaric acid, and caffeic acid). Total chlorogenic acids (classified here as polyols according to the KEGG database), and trigonelline (pyridine alkaloid) were classified separately.

This analysis indicated that PRO coffees from Cerrado have higher levels of trigonelline than PRO coffees from Matas de Minas (Figure 3D; $p = 0.0131$), but there was no significant difference for NAT coffees ($p = 0.4438$). There were no significant differences in cinnamic acid derivatives levels between regions or types of processing (Figure 3A). However, for chlorogenic acid derivatives and xanthines (Figure 3B and 3C, respectively), two-way ANOVA revealed an interaction between the factors “processing” ($p = 0.0427$) and “region of origin” ($p = 0.0315$), indicating that this interaction can impact the final chlorogenic acid derivatives and xanthines profiles.

3.3 Effects of post-harvest processing on the chemical composition

To enhance product quality and market value, farmers frequently adopt alternative coffee post-harvesting processing techniques. A common method involves de-pulping, in which the skin and pulp are removed, leaving behind the mucilage-covered parchment that encases the seed. In the sequence, the beans are subjected to controlled drying. This process fosters fermentation by various microbial groups, which can alter the bean's endogenous metabolism, thereby modifying its final chemical profile and flavor characteristics. Such green beans, when roasted, are known to produce a beverage with a distinctive profile of low acidity, a sweet taste, and a characteristic aroma (Fernandes et al., 2014; Várady et al., 2022, Velásquez & Banchón, 2023). On the opposite, in natural processing, the coffee cherries are usually spread outdoors, under the sun on cement or brick surfaces from ten days to three weeks. Non-uniform drying may occur depending on the thickness of the layer of coffee cherries on the surface. Over-dried cherries can become brittle, while under-dried ones are highly susceptible to fungal and bacterial growth. Rain contributes to under-drying, increasing the risk of non-controlled secondary fermentation due to the highly hygroscopic mucilage that remains in the coffee beans. Furthermore, the cleaning conditions of the drying surface can affect the final taste of the drink due to the transfer of impurities to the beans (Haile & Kang, 2020).

A closer look at the impact of the de-pulping process on the coffee beans' bioactive compounds can also be observed in Figure 3. A statistically significant increase in the chlorogenic acid derivatives and xanthines was noted only in the samples from Matas de Minas, post-de-pulping. Interestingly, NAT samples from Matas de Minas also showed lower amounts of all compound classes, suggesting that the process of de-pulping can, indeed, increase the concentration of biomarkers in samples with initially lower amounts of phenolics and alkaloids.

Altogether, taking into account the samples from all regions, statistically significant results were observed only for chlorogenic acid (CGA) derivatives. Pulped coffees displayed 5.8% more CGA derivatives compared to unprocessed counterparts. Such compounds and other polyphenols such as gallic acid, feruloyl quinic, and dicaffeoyl quinic acid derivatives, and the lactones of CGA are abundant in coffee. These compounds are not only pivotal to the unique taste profile of the beverage but they are also recognized for their potential health benefits, including the prevention of various chronic and degenerative diseases (Várady et al., 2022; Febrianto & Zhu, 2023). However, during roasting, it is known that part of CGAs

decomposes to form quinic and caffeic acids, which are associated with astringency, bitterness, and acidity (Ogutu et al., 2022).

In addition, regardless of region, de-pulped coffees also showed significantly higher sensory scores when compared to natural coffees (Figure 4). Compared to natural beans, de-pulped coffee showed a 1.5% significantly higher average sensory score ($p = 0.0238$), corroborating that the post-harvesting practices sought higher grading scores. Therefore, the observed results suggest that the de-pulping processing can contribute to improving the sensory score and increasing the content of health-beneficial compounds.

3.4 Correlation between processing type, biomarkers composition, and sensory evaluation

Currently, the industry lacks a definitive classification system that correlates coffee bean quality with its chemical composition and biological activity. Trigonelline, for instance, a major compound in coffee, can contribute positively to the sensory evaluation. During the roasting process, trigonelline decomposes into nicotinic acid, pyrroles, and pyridine, which impart nutty and toasted notes, and it is associated with desirable aroma attributes (Ogutu et al., 2022). On the other hand, caffeine content, while commonly used to distinguish between *C. arabica* (lower caffeine) and *C. robusta* (higher caffeine), is not a direct indicator of bean quality (Febrianto & Zhu, 2023). However, it is already known that high-quality coffee beans present lower caffeine and chlorogenic acid content but higher sucrose, which is associated with a better taste profile (Barbosa et al., 2019).

To contribute to this topic and to critically assess the results of the present study, multivariate logistic regression (MLR) analysis was performed to establish correlations among the processing types, the several graded sensory attributes including clean cup, sweetness, flavor, acidity, body, balance, aftertaste, and overall perception, and each one of the bioactive chemical markers.

The MLR model pointed out that pulped coffees were statistically more associated with a higher sweetness grade, while naturally processed beans tended to exhibit a higher acidity level. Thus, the sensory predictors sweetness and acidity attributes were statistically related to the post-harvesting method ($p < 0.04$ and $p < 0.03$, respectively), suggesting that these particular attributes significantly contribute to the sensory differentiation of the two processing types. The correlation matrix further supported the role of sweetness as a significant individual predictor (Figure 5) and Student's t-tests confirmed pulped coffees exhibited significantly higher sweetness scores compared to natural coffees ($p < 0.0033$) (Figure 5B). These results also align with previously published data in which high levels of sucrose and lower acidity correlated to high-grade green coffee beans (Kwon et al., 2015; Hall, Trevisan & De Vos, 2022).

Regarding the chemical composition, the MLR model indicated processed beans with significantly higher concentrations of theobromine ($p < 0.044$) and *p*-coumaric acid ($p < 0.0026$), while natural coffees showed significantly higher levels of ferulic acid ($p < 0.004$). These findings were corroborated by Mann-Whitney tests, which verified a significant

increase in *p*-coumaric acid content in de-pulped compared to the natural coffee ($p < 0.0090$) (Figure 5B). The influence of theobromine as well as theophylline and other less common alkaloids remains underexplored on the quality of coffee, especially after roasting (Hall, Trevisan & De Vos, 2022). The sensory role of *p*-coumaric acid and ferulic acid follows the same path and not much is known about its relevance in the sensory quality. Together with other cinnamic acid derivatives, *p*-coumaric acid represents less than 1% of the total phenolic compounds in coffee beans (Hall, Trevisan & De Vos, 2022). Despite the lack of information concerning these two predictors, univariate analyses provided further support for these compound differences between processing methods. Spearman's correlation showed inverse relationships between ferulic acid content and overall sensory score in both NAT and PRO coffees, but stronger for PRO (Figure 5 and Table S5). Additionally, ferulic acid levels were lower in PRO beans ($p < 0.0199$) (Figure 5B). Therefore, the results together suggest that the processing method can affect the concentration of these compounds, with a lower ferulic acid content being associated with higher quality in the de-pulped beans.

Major compounds such as caffeine and chlorogenic acid derivatives levels did not differ significantly between processing methods according to the MLR model. However, in PRO coffees, lower caffeine, and theophylline were linked to improved sensory aspects, as evidenced by the correlation matrix analysis. Also, the negative correlation between the sensory score and both xanthines confirmed this observation (Figure 5 and Table S5).

The fitted MLR model also provided evidence that higher growing altitudes significantly predict pulped rather than non-processed coffees based on the estimated odds ratio of 0.980 (95% CI: 0.964 - 0.991; $p < 0.0038$). In the same way, the correlation matrix (Figure 5) corroborates the MLR results, indicating a strong positive association between altitude and scores in sensory aspects, specifically for PRO coffees. Finally, the fitted MLR model incorporating factors such as altitude, phytochemical, and sensory predictors demonstrated an exceptional predictive accuracy, with an area under the ROC curve of 0.958 (95% CI: 0.917 to 0.999). It accurately classified 91% of cases in distinguishing between natural and de-pulped processing based on a combined analysis of altitude as a geographical aspect, concentrations of ferulic and *p*-coumaric acids, and sensory evaluations of sweetness and acidity (Figure S3).

4. Conclusions

The present study contributed to the elucidation of the complex interplay between the geographical origin, post-harvest processing, phytochemical profile, and the resultant sensory attributes of Brazilian specialty coffee beans from Minas Gerais State. The findings underscore the dual influence of both origin and post-harvest processing methods on the bean's chemical and sensory quality. Based on the bioactive chemical profile, Cerrado samples showed a tendency to be separated from the remaining Atlantic Forest regions. Although with an evident overlapping, this finding sheds light on the possibility of employing target quantitative analysis in future coffee geographical designation endeavors.

Regarding the post-harvest processing, the de-pulping significantly increased levels of beneficial phytochemicals like chlorogenic acid derivatives. Besides, processed beans showed a 1.5% higher average sensory score versus natural beans, correlating with the 5.8% increase in chlorogenic acid derivatives content. However, phenolic compounds, beyond their valuable health-beneficial properties, also contribute to the bitterness and astringency of coffee, which can adversely affect sensory quality. The challenge lies in developing processing methods that strike a balance between enhancing the biological properties and maintaining desirable sensory attributes of the coffee beans.

The MLR analysis supported by the univariate statistical analysis points out altitude, ferulic acid, *p*-coumaric acid, and sweetness as important predictors distinguishing specialty coffee beans that are either non-processed or have undergone de-pulping. These statistical methods provide robust evidence that post-harvest processing induces changes in coffee phytochemistry, which in turn influences their sensory appeal and overall quality.

In essence, the approach of targeted metabolite profiling can complement sensory evaluation for specialty coffee authentication and quality assessment. Both the geographical origin and the post-harvest techniques are instrumental in shaping the chemical and flavor profiles of green coffee beans. Further research with expanded sample sizes and increased metabolite coverage will certainly contribute to validating these relationships between geographical origin, processing methods, sensory profiles, and chemical composition.

CRedit authorship contribution statement

Heloisa Tieghi: Investigation, Methodologies, Data curation

Luana de Almeida Pereira: Investigation, Methodologies

Gabriel Silva Viana: Investigation, Methodologies, Visualization, Software, Writing – review & editing

Albert Katchborian: Investigation, Methodologies, Visualization, Software, Writing – review & editing

Derielsen Brandão Santana: Methodology, Visualization, Software

Ronaldo Luiz Mincato: Project administration

Danielle Ferreira Dias: Funding acquisition, Resources

Daniela Aparecida Chagas de Paula: Funding acquisition, Resources, Writing – review & editing

Marisi Gomes Soares: Funding acquisition, Resources

Willem Guilherme de Araújo: Project administration, Resources

Paula Carolina Pires Bueno: Conceptualization, Formal analysis, Investigation, Project administration, Supervision, Funding acquisition, Writing – original draft, Writing – review & editing, Supervision

Declaration of competing interest

The authors declare no conflicts of interest.

Acknowledgment

This work was supported by Emater-MG, the National Council for Scientific and Technological Development (CNPq, finance codes 316204/2021-8 and 406837/2021-0), Coordination for the Improvement of Higher Education Personnel (CAPES, finance code 001) and Minas Gerais State Research Foundation (FAPEMIG, finance codes APQ-02353-17 and APQ-00207-18). Heloisa Tieghi received an institutional fellowship from the Federal University of Alfenas (PIBICT/FAPEMIG 017/2021 PRPPG and 011/2022 PRPPG). The authors also thank Dr. Joachim Kopka from the Max Planck Institute of Plant Molecular Physiology for providing the authentic standards used in the study.

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Figures

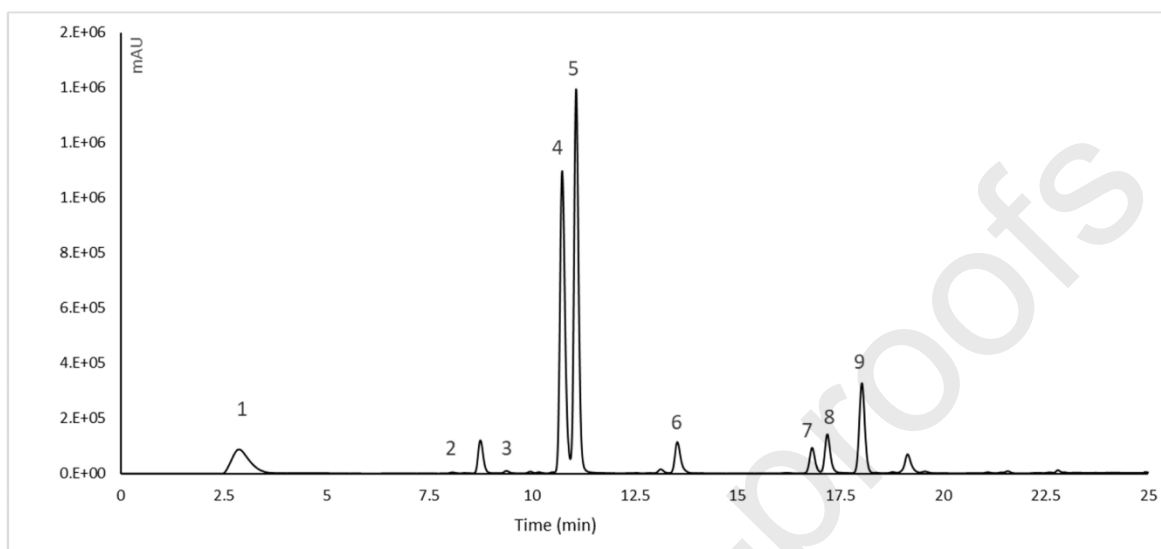


Figure 1. Chromatographic profile of a representative sample of specialty green coffee beans showing the major compounds quantified in the present study. (1) trigonelline, (2) theobromine, (3) theophylline, (4) caffeine, (5) chlorogenic acid derivatives, (6) caffeic acid, (7) ferulic acid, (8) *p*-coumaric acid, and (9) naringenin (internal standard). The chromatogram was plotted at 275 nm.

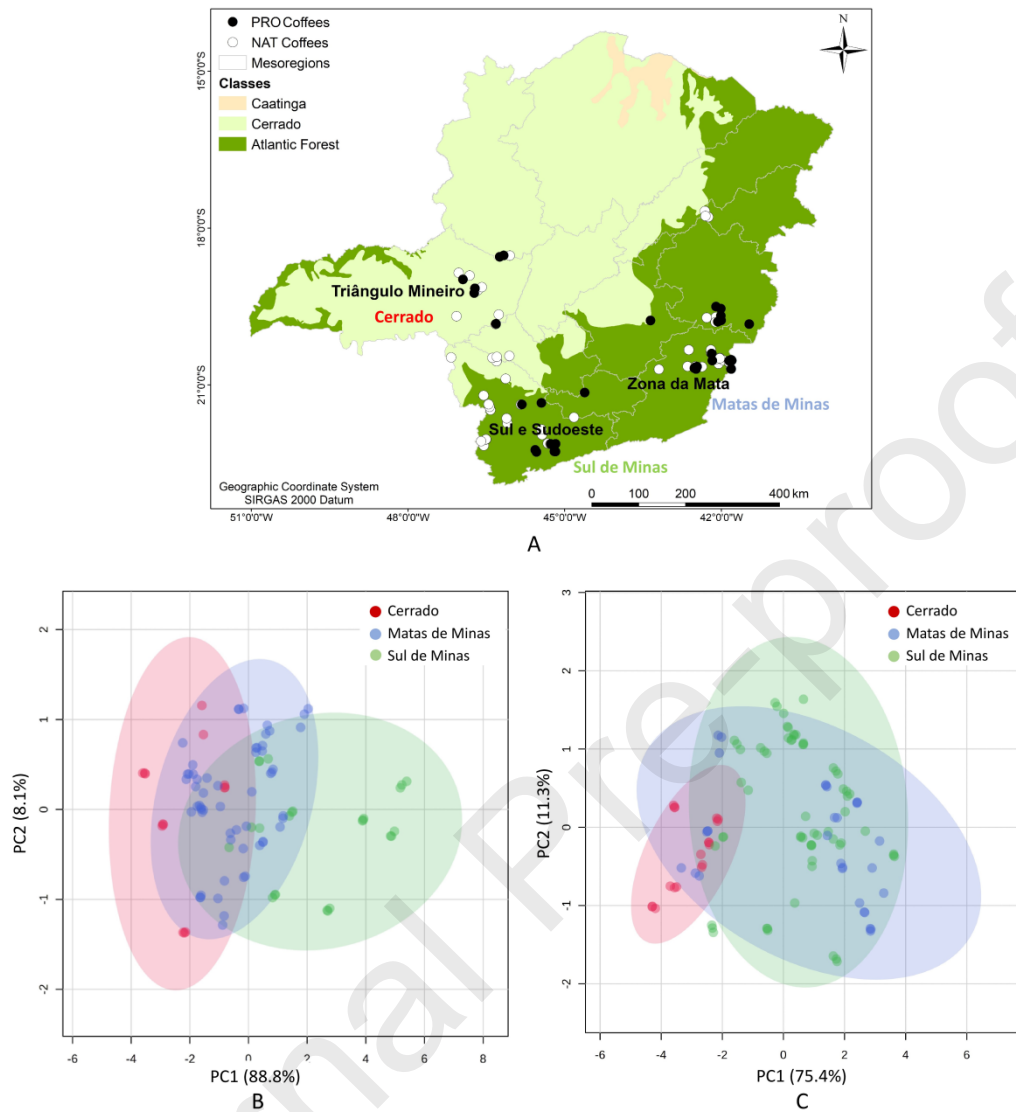


Figure 2. (A) Map of Minas Gerais state showing the coffee samples' distribution along the three main regions. White dots = NAT (non-processed coffee beans); Black dots = PRO (processed coffee beans). (B) PCA score plots illustrating the clustering patterns between non-processed (NAT) coffee beans from Cerrado, Matas de Minas, and Sul de Minas. (C) PCA score plots illustrating the clustering patterns between processed (PRO) coffee beans from Cerrado, Matas de Minas, and Sul de Minas. Ellipses represent a 95% confidence interval for each group. Note: the map of phytogeographic domains was elaborated based on land use and occupation data extracted from Collection 8 of the MapBiomas platform (Supplementary information Text S1).

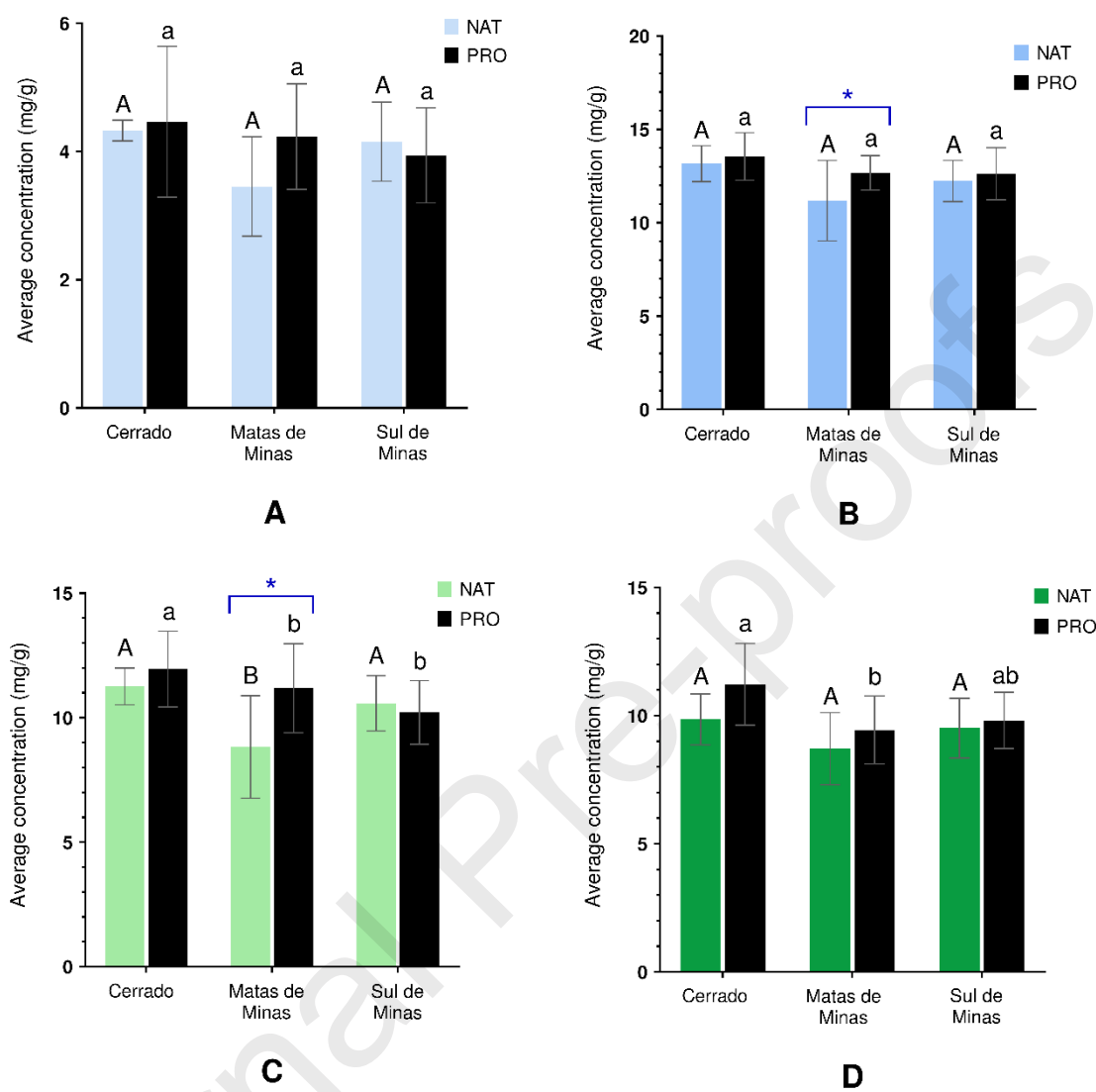


Figure 3. Influence of processing type and geographical location on the average concentration of enriched biomarkers in green coffee beans. (A) Cinnamic acid derivatives; (B) Chlorogenic acid derivatives; (C) Xanthines; (D) Trigonelline. The * represents significant differences between the processing types. The error bars represent the variability of the measurements given by the standard deviation (n = 39). The different upper case letters A and B above the columns represent significant differences in non-processed samples (NAT) between different locations. The different lowercase letters a and b represent significant differences in processed samples (PRO) between different locations. Results were calculated using the Tukey test ($p < 0.05$) after two-way ANOVA.

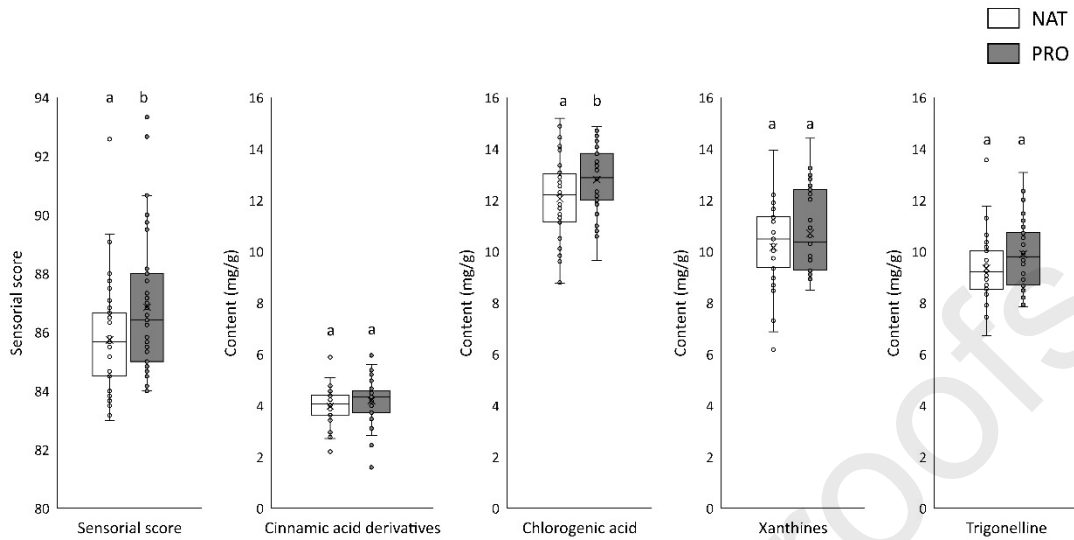


Figure 4. Box plots showing the distribution of sensory scores and biomarkers contents within all 39 non-processed coffee beans (NAT) and all 39 processed coffee beans (PRO). The different lowercase letters a and b represent significant differences, accessed by Student's t-test ($p < 0.05$).

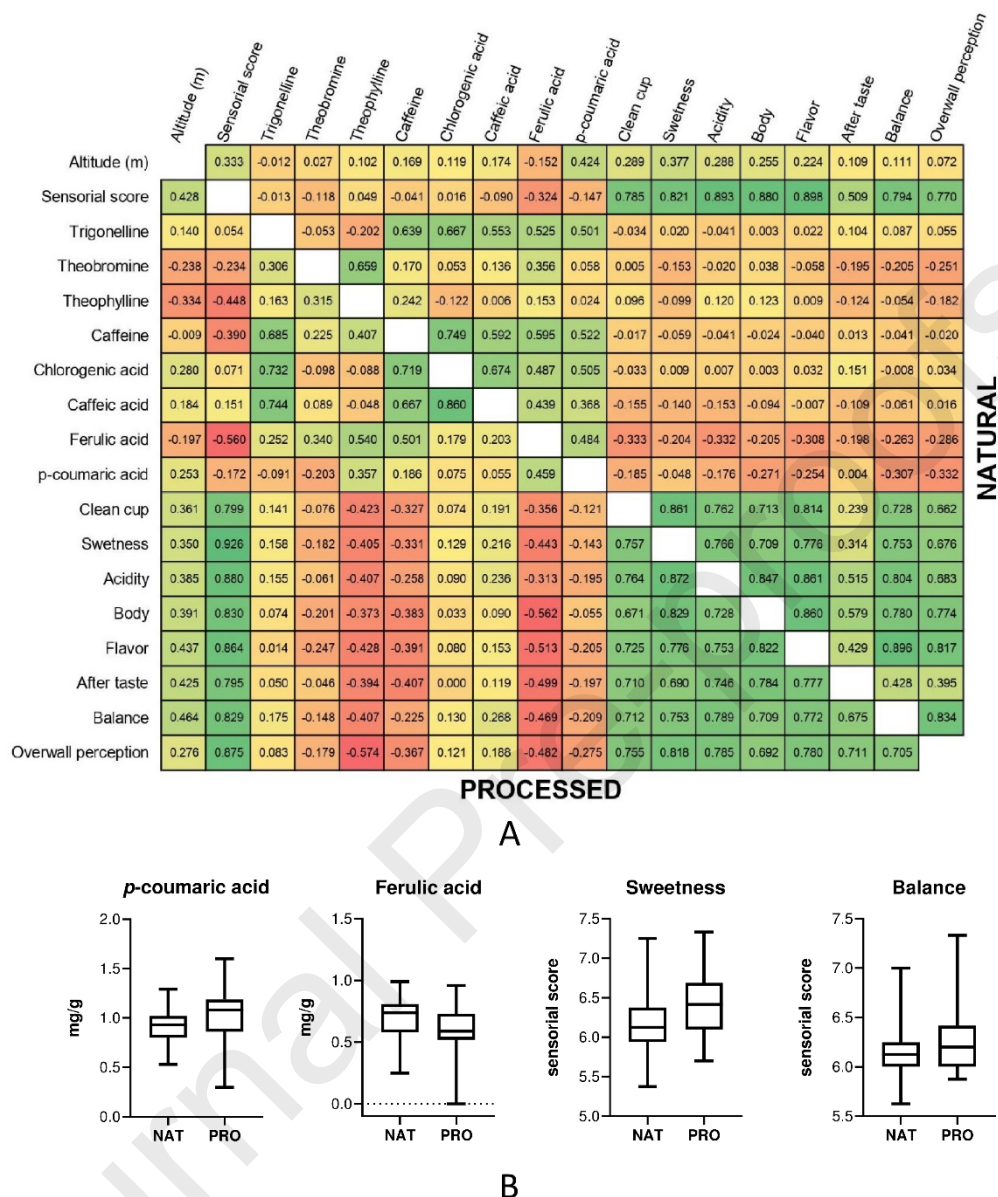


Figure 5. (A) Heatmap generated from Spearman's rank correlation analysis, illustrating the interrelationships among various chemical compounds, sensory attributes, and the altitude of coffee bean cultivation. Each cell within the matrix denotes the correlation coefficient between pairs of variables, with color intensity and direction (positive or negative) reflecting the strength and nature of the correlation. (B) Mann-Whitney test results show the differences of *p*-coumaric (*p*-value = 0.0090), ferulic acid (*p*-value = 0.0199), and balance (*p*-value = 0.0454) between NAT and PRO samples. Sweetness was accessed by the Student's *t*-test (*p*-value = 0.0033).

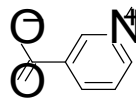
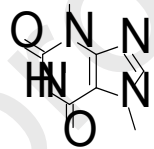
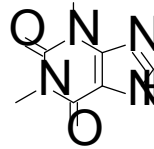
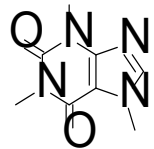
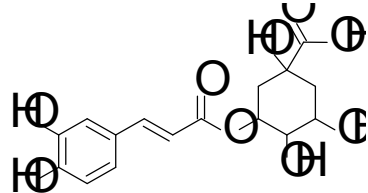
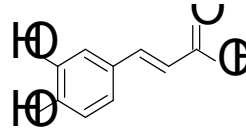
Tables

Table 1. Geographical regions of origin, altitude range, type of processing, and number of samples of the green coffee beans analyzed in this study.

Region	Altitude	Type of processing	Number of samples
Cerrado	930-1262	NAT	5
	980-1230	PRO	7
Sul de Minas	850-1370	NAT	23
	810-1319	PRO	10
Matas de Minas	790-1340	NAT	11
	700-1336	PRO	22

NAT = natural coffee beans; PRO = processed coffee beans.

Table 2. Compounds' names, retention times, UV maximum wavelength, and chemical structures of the biomarkers analyzed in the specialty green coffee beans.

#	Compound	Retention time (min)	$\lambda_{\max}$ (nm)	Structure
1	Trigonelline	2.87	265	
2	Theobromine	8.05	272	
3	Theophylline	9.37	275	
4	Caffeine	11.07	271	
5	Chlorogenic acid derivatives (expressed in 3-CQA)	10.73	325	
6	Caffeic acid	13.53	322	

7	Ferulic acid	16.82	322	<chem>CC(=O)/C=C/c1ccc(O)c(OC)c1</chem>
8	<i>p</i> -coumaric acid	17.18	326	<chem>C=C/c1ccc(O)cc1</chem>

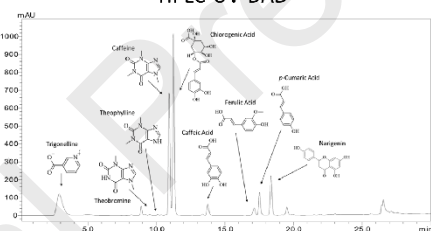
Effects of geographical origin and post-harvesting processing on the bioactive compounds and sensory quality of Brazilian specialty coffee beans

Harvesting and processing

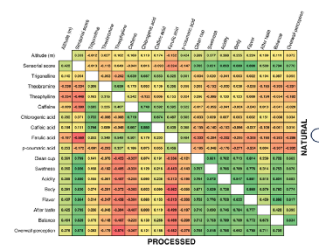


Data acquisition

HPLC-UV-DAD



Correlation analysis



Outcome

- Targeted metabolite profiling can complement sensory analysis for specialty coffee grading and support origin authentication
- Post-harvesting processing can increase the concentrations of health-beneficial compounds

Highlights

Target metabolite profiling supported the characterization of specialty coffee beans

The geographical origin influenced specialty coffee's chemical profile and quality

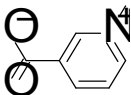
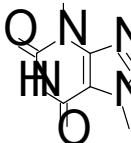
Post-harvesting processing can enhance the content of health-beneficial compounds

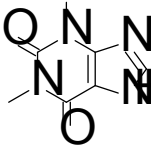
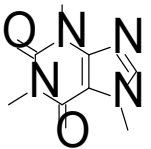
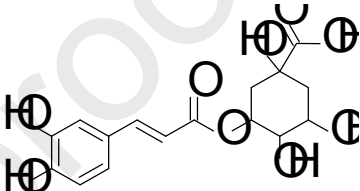
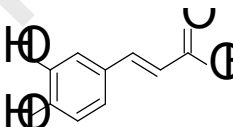
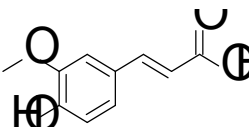
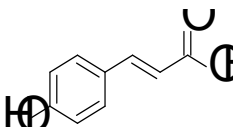
Table 1. Geographical regions of origin, altitude range, type of processing and number of samples of the green coffee beans analyzed in this study.

Region	Altitude	Type of processing	Number of samples
Cerrado	930-1262	NAT	5
	980-1230	PRO	7
Sul de Minas	850-1370	NAT	23
	810-1319	PRO	10
Matas de Minas	790-1340	NAT	11
	700-1336	PRO	22

NAT = natural coffee beans; PRO = processed coffee beans.

Table 2. Compounds' names, retention times, UV maximum wavelength, and chemical structures of the biomarkers analyzed in the specialty green coffee beans.

#	Compound	Retention time (min)	$\lambda_{\max}$ (nm)	Structure
1	Trigonelline	2.87	265	
2	Theobromine	8.05	272	

3	Theophylline	9.37	275	
4	Caffeine	11.07	271	
5	Chlorogenic acid derivatives (expressed in 3-CQA)	10.73	325	
6	Caffeic acid	13.53	322	
7	Ferulic acid	16.82	322	
8	<i>p</i> -coumaric acid	17.18	326	

Declaration of competing interest

The authors declare no conflicts of interest